

REFERENSI PERINTAH

Linux System Administration

Praktikum PATI - Mini Server Linux

April 2026

1. Boot & Service Management

1.1 Identifikasi Sistem

Perintah	Fungsi
uname -r	Menampilkan versi kernel yang sedang berjalan
uname -a	Info lengkap kernel (hostname, arsitektur, tanggal build)
grub-install --version	Menampilkan versi GRUB boot loader
cat /etc/os-release	Info distribusi OS (nama, versi, ID)
hostnamectl	Info hostname, OS, kernel, arsitektur secara ringkas
lsb_release -a	Info distribusi Linux (jika tersedia)

1.2 Manajemen Service (systemctl)

Perintah	Fungsi
systemctl status <service>	Cek status service (active/inactive/failed)
systemctl start <service>	Menjalankan service
systemctl stop <service>	Menghentikan service
systemctl restart <service>	Restart service (stop + start)
systemctl reload <service>	Reload konfigurasi tanpa restart
systemctl enable <service>	Aktifkan service saat boot
systemctl disable <service>	Nonaktifkan service saat boot
systemctl enable --now <svc>	Enable + langsung start
systemctl disable --now <svc>	Disable + langsung stop
systemctl is-enabled <service>	Cek apakah service di-enable
systemctl is-active <service>	Cek apakah service sedang berjalan
systemctl list-units --type=service	Daftar semua service yang loaded
systemctl list-unit-files --type=service	Daftar semua unit file service
systemctl mask <service>	Blokir service agar tidak bisa di-start
systemctl unmask <service>	Lepas blokir service

1.3 Journal & Log

Perintah	Fungsi
journalctl -u <service>	Log untuk service tertentu
journalctl -u <service> -f	Follow log real-time
journalctl -b	Log sejak boot terakhir
journalctl --since '1 hour ago'	Log 1 jam terakhir
journalctl -p err	Hanya tampilkan error
dmesg	Kernel ring buffer (log hardware/boot)
dmesg tail -20	20 pesan kernel terakhir

1.4 Custom Service File

Lokasi: `/etc/systemd/system/hello-boot.service`

Perintah	Fungsi
<code>systemctl daemon-reload</code>	Reload setelah edit/buat unit file baru
<code>systemctl enable hello-boot</code>	Aktifkan custom service saat boot
<code>systemctl start hello-boot</code>	Jalankan manual
<code>systemctl status hello-boot</code>	Cek status custom service

2. Manajemen Disk & Partisi

2.1 Informasi Disk

Perintah	Fungsi
lsblk	Tampilkan struktur block device (disk, partisi, mount)
lsblk -f	Tampilkan + filesystem & UUID
fdisk -l	Detail partisi semua disk
blkid	UUID dan tipe filesystem tiap partisi
df -h	Penggunaan disk per mount point (human-readable)
df -i	Penggunaan inode per mount point
du -sh /path	Ukuran total sebuah direktori
du -sh /*	Ukuran tiap direktori di root

2.2 Partisi & Format

Perintah	Fungsi
sudo fdisk /dev/sdb	Partisi disk sdb (interactive)
sudo parted /dev/sdb mklabel gpt	Buat GPT label pada disk
sudo parted /dev/sdb mkpart primary ext4 40% 100%	Buat partisi penuh
sudo mkfs.ext4 /dev/sdb1	Format partisi sebagai ext4
sudo mkfs.xfs /dev/sdb1	Format partisi sebagai XFS

2.3 Mount & fstab

Perintah	Fungsi
sudo mount /dev/sdb1 /data	Mount partisi ke /data
sudo umount /data	Unmount partisi
mount grep sdb	Cek status mount disk sdb
cat /etc/fstab	Lihat konfigurasi mount permanen
sudo nano /etc/fstab	Edit fstab untuk auto-mount
sudo mount -a	Mount ulang semua entry di fstab

3. Keamanan Data (Backup & Recovery)

3.1 rsync (Backup)

Perintah	Fungsi
<code>rsync -a /source/ /dest/</code>	Backup arsip (permissions, timestamps)
<code>rsync -av /source/ /dest/</code>	Backup verbose (tampilkan file)
<code>rsync -a --delete /src/ /dst/</code>	Sinkronisasi mirror (hapus file extra di dest)
<code>rsync -a /data/ /backup/</code>	Backup /data ke /backup
<code>rsync -a /backup/share/ /data/share/</code>	Restore folder share dari backup

3.2 Cron (Penjadwalan)

Perintah	Fungsi
<code>crontab -e</code>	Edit cron untuk user saat ini
<code>sudo crontab -e</code>	Edit cron untuk root
<code>crontab -l</code>	Lihat daftar cron user saat ini
<code>sudo crontab -l -u root</code>	Lihat cron root
<code>sudo crontab -r</code>	Hapus semua cron root

Format Cron: menit jam hari bulan hari-minggu perintah

Perintah	Fungsi
<code>0 2 * * *</code>	Setiap hari jam 02:00
<code>* /5 * * * *</code>	Setiap 5 menit
<code>0 0 * * 0</code>	Setiap Minggu jam 00:00
<code>0 3 1 * *</code>	Tanggal 1 setiap bulan jam 03:00

3.3 Simulasi Kegagalan & Recovery

Perintah	Fungsi
<code>sudo rm /data/share/Hi.txt</code>	Simulasi kehilangan file
<code>sudo rsync -a /backup/share/ /data/share/</code>	Restore dari backup
<code>cat /data/share/Hi.txt</code>	Verifikasi file berhasil dikembalikan
<code>ls -l /data/share/</code>	Cek isi direktori setelah restore

4. Monitoring Sistem

4.1 Monitoring Interaktif

Perintah	Fungsi
htop	Monitor proses interaktif (CPU, RAM, per-proses)
top	Monitor proses (bawaan, lebih sederhana)
top -bn1 head -20	Snapshot non-interaktif top

4.2 Monitoring Resource

Perintah	Fungsi
free -h	Penggunaan RAM dan swap
vmstat 1 5	Statistik VM: CPU, memory, IO (5 iterasi)
iostat	Statistik I/O disk dan CPU
sar -u 1 5	CPU usage per detik (5 kali)
sar -r 1 5	Memory usage per detik
uptime	Load average dan uptime
nproc	Jumlah CPU core
lscpu	Info detail CPU

4.3 Stress Testing

Perintah	Fungsi
stress --cpu 2 --timeout 30	Bebankan 2 CPU selama 30 detik
stress --cpu 4 --timeout 60	Bebankan 4 CPU selama 60 detik
stress --vm 1 --vm-bytes 256M --timeout 30	Stress test RAM 256MB
stress --io 2 --timeout 30	Stress test I/O 2 worker

5. Update Sistem & Keamanan

5.1 Update Paket (APT)

Perintah	Fungsi
<code>sudo apt update</code>	Refresh daftar paket dari repository
<code>sudo apt upgrade -y</code>	Upgrade semua paket yang tersedia
<code>sudo apt full-upgrade -y</code>	Upgrade + hapus paket lama jika perlu
<code>sudo apt autoremove -y</code>	Hapus paket yang tidak lagi diperlukan
<code>sudo apt install <paket></code>	Install paket baru
<code>sudo apt remove <paket></code>	Hapus paket
<code>apt list --installed</code>	Daftar paket yang terinstall
<code>apt list --upgradable</code>	Daftar paket yang bisa di-upgrade
<code>dpkg -l grep <keyword></code>	Cari paket terinstall by keyword

5.2 Firewall (UFW)

Perintah	Fungsi
<code>sudo ufw enable</code>	Aktifkan firewall
<code>sudo ufw disable</code>	Nonaktifkan firewall
<code>sudo ufw status</code>	Status dan rules aktif
<code>sudo ufw status verbose</code>	Status detail dengan default policies
<code>sudo ufw allow 22/tcp</code>	Izinkan SSH
<code>sudo ufw allow Samba</code>	Izinkan Samba (profil aplikasi)
<code>sudo ufw deny 23/tcp</code>	Blokir Telnet
<code>sudo ufw delete allow 22/tcp</code>	Hapus rule allow SSH
<code>sudo ufw reset</code>	Reset semua rules ke default
<code>sudo ufw app list</code>	Daftar profil aplikasi yang tersedia

5.3 Timezone

Perintah	Fungsi
<code>timedatectl</code>	Status waktu dan timezone saat ini
<code>timedatectl list-timezones</code>	Daftar semua timezone
<code>sudo timedatectl set-timezone Asia/Jakarta</code>	Set timezone ke WIB
<code>date</code>	Tampilkan tanggal dan waktu saat ini

6. Samba File Sharing

6.1 Instalasi & Service

Perintah	Fungsi
<code>sudo apt install samba -y</code>	Install Samba
<code>systemctl status smbd</code>	Cek status Samba daemon
<code>systemctl restart smbd</code>	Restart Samba
<code>testparm</code>	Validasi konfigurasi smb.conf

6.2 User & Password

Perintah	Fungsi
<code>sudo smbpasswd -a adminuser</code>	Tambah/set password Samba user
<code>sudo smbpasswd -x adminuser</code>	Hapus Samba user
<code>sudo pdbedit -L</code>	Daftar Samba users

6.3 Konfigurasi Share

File: **/etc/samba/smb.conf**

Perintah	Fungsi
<code>sudo nano /etc/samba/smb.conf</code>	Edit konfigurasi Samba
<code>testparm -s</code>	Validasi dan tampilkan config

6.4 Akses dari Windows

Perintah	Fungsi
<code>\\192.168.56.101</code>	Akses via File Explorer (address bar)
<code>net use Z: \\192.168.56.101\PraktikumShare</code>	Map network drive via CMD
<code>net use Z: /delete</code>	Hapus mapped drive

6.5 Hak Akses File

Perintah	Fungsi
<code>chmod 755 /data/share</code>	Set permissions (rwxr-xr-x)
<code>chown adminuser:adminuser /data/share</code>	Set ownership
<code>ls -la /data/share/</code>	Lihat permissions detail

7. Ansible (Infrastructure-as-Code)

7.1 Perintah Dasar

Perintah	Fungsi
<code>ansible --version</code>	Cek versi Ansible
<code>ansible all -m ping -i inventory.ini</code>	Test koneksi ke semua host
<code>ansible-playbook setup.yml -i inventory</code>	Jalankan playbook
<code>ansible-playbook setup.yml -i inventory -vvv</code>	Jalankan dengan verbose
<code>ansible-playbook setup.yml --check</code>	Dry run (tidak eksekusi)
<code>ansible-playbook setup.yml --tags disk</code>	Jalankan hanya tag tertentu

7.2 Inventory (inventory.ini)

```
ansible_host=192.168.56.101
ansible_user=adminuser
ansible_ssh_pass=testPraktikum
ansible_become_pass=testPraktikum
```



```
proto: tcp

- name: Allow Samba
community.general.ufw:
    rule: allow
    name: Samba

- name: Deny Telnet (port 23)
community.general.ufw:
    rule: deny
    port: '23'
    proto: tcp

- name: Enable UFW
community.general.ufw:
    state: enabled
    default: deny

# ████████████████████████████████████████████████████████████████
# 5. DISK PARTISI (/data)
# ████████████████████████████████████████████████████████████████

- name: Create partition on /dev/sdb
community.general.parted:
    device: /dev/sdb
    number: 1
    state: present
    label: gpt
    part_start: 0%
    part_end: 100%

- name: Format /dev/sdb1 as ext4
community.general.filesystem:
    fstype: ext4
    dev: /dev/sdb1

- name: Create /data mount point
file:
    path: /data
    state: directory

- name: Mount /dev/sdb1 to /data
mount:
    path: /data
    src: /dev/sdb1
    fstype: ext4
    state: mounted
    opts: defaults
    dump: '0'
    passno: '2'

# ████████████████████████████████████████████████████████████████
# 6. DISABLE MULTIPATHD
# ████████████████████████████████████████████████████████████████

- name: Disable multipathd service
systemd:
    name: multipathd
    enabled: no
    state: stopped
ignore_errors: yes

# ████████████████████████████████████████████████████████████████
# 7. CUSTOM STARTUP SERVICE
# ████████████████████████████████████████████████████████████████

- name: Create hello-boot.service unit file
copy:
    dest: /etc/systemd/system/hello-boot.service
    content: |
        [Unit]
        Description=Hello Boot - Custom Startup Service
        After=network.target

        [Service]
```

```
[Install]
WantedBy=multi-user.target
```

```
# ████████████████████████████████████████████████████████████  
# 8. BACKUP (RSYNC + CRON)  
# ████████████████████████████████████████████████████████████
```

```
# ████████████████████████████████████████████████████████████  
# 9. SAMBA FILE SHARING  
# ████████████████████████████████████████████████████████████
```

```
- name: Restart Samba
  systemd:
    name: smbd
    state: restarted
    enabled: yes
```

8.3 hello-boot.service

Lokasi: **/etc/systemd/system/hello-boot.service**

Custom systemd unit file untuk mencatat pesan saat boot.

```
[Unit]
Description=Hello Boot - Custom Startup Service
After=network.target

[Service]
Type=oneshot
ExecStart=/bin/echo "Hello! Server booted at $(date)"
StandardOutput=journal

[Install]
WantedBy=multi-user.target
```

8.4 smb.conf (Samba Share Section)

Lokasi: **/etc/samba/smb.conf** (ditambahkan di akhir file)

```
[PraktikumShare]
comment = Praktikum File Share
path = /data/share
browseable = yes
read only = no
valid users = adminuser
create mask = 0644
directory mask = 0755
```

8.5 /etc/fstab (Entry Disk Data)

Entry yang ditambahkan oleh Ansible untuk auto-mount disk data saat boot.

```
/dev/sdb1    /data    ext4    defaults    0    2
```

8.6 Crontab Root (Backup Harian)

Entry cron yang ditambahkan oleh Ansible untuk backup otomatis.

```
# Ansible: Daily data backup
0 2 * * * rsync -a /data/ /backup/ > /var/log/data_backup.log 2>&1
```

8.7 Penjelasan Struktur setup.yml

Bagian	Fungsi
hosts: servers	Target host dari inventory.ini
become: yes	Jalankan sebagai root (sudo)
apt: update_cache	Update daftar paket (apt update)
apt: upgrade: dist	Upgrade semua paket
community.general.timezone	Set timezone sistem
apt: name: [htop, sysstat, stress]	Install tools monitoring
community.general.ufw	Konfigurasi firewall rules
community.general.parted	Buat partisi disk baru
community.general.filesystem	Format partisi (mkfs)
mount:	Mount disk + tambahkan ke fstab

systemd: enabled: no	Disable service (multipathd)
copy: content:	Buat file baru (hello-boot.service)
systemd: daemon_reload: yes	Reload systemd setelah buat unit
file: state: directory	Buat direktori (/backup, /data/share)
cron:	Tambahkan cron job otomatis
blockinfile:	Tambahkan blok teks ke file (smb.conf)
systemd: state: restarted	Restart service (smbd)